AECCS Midterm Presentation

Assessing the Effectiveness of Cookie Consent Systems through automated measurement, tracker analysis, dark-pattern detection, and PET evaluation.

CS475 Privacy-Enhancing Technologies

Problem Statement

- Cookie banners are supposed to provide meaningful user control over tracking.
- In practice, many banners use dark patterns or fail to enforce rejection choices.
- The project asks whether consent systems protect privacy and whether PETs reduce exposure when they do not.

Related Work

- Nouwens et al.: dark patterns remain widespread after GDPR.
- Matte et al.: many banners do not technically respect user choice.
- Bollinger et al.: automated cookie-consent analysis can be scaled.
- GDPR and CNIL guidance define the baseline consent expectations.

Implemented System

Component	Status
Playwright crawler	Implemented
Tracker classifier	Implemented
Dark-pattern detector	Implemented
Compliance scoring	Implemented
Aggregate metrics	Implemented
Browser PET evaluation	Implemented
CMP analysis / DP reporting / reporting	Implemented

Current Milestone

- Mock and real datasets are separated.
- Artifacts now carry provenance metadata and run IDs.
- Report generation rejects mixed processed inputs.
- Validation tests cover tracker classification, dark patterns, scoring, PET skip behavior, and reporting contracts.

Current percentages in the repository are synthetic demo values, not final study findings.

Next Steps

- Run the 100-site real study list under `data/real/`.
- Execute the PET experiments on a real subset or the full list.
- Regenerate figures, report outputs, and submission artifacts from real data only.